

Cancer-Fighting Lettuce: Engineering Betalain Biosynthesis in *Lactuca sativa* through CYP76AD1, DODA, and UGT73A6

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Introduction

Betalains are red and yellow pigments derived from tyrosine, found in plants of the Caryophyllales order. Renowned for their antioxidant, anti-lipidemic, antimicrobial, anti-inflammatory & anticancer properties, betalains have drawn significant attention for their potential health and pharmaceutical benefits. Betanin, a specific type of betalain, demonstrates promising antitumor activity by inhibiting pathways such as TPA, DMBA, NOR-1, and DEN. Its biological effects include reducing cell proliferation, promoting apoptosis and autophagy, and decreasing tumor angiogenesis. Up until now, the heterologous engineering of the betalain biosynthesis pathway (BBP) has led to the BBP reconstitution and subsequent biofortification in solely tomatoes, potatoes, carrots, rice, yeast, and silk moths. However, in this research, I report the novel optimization and engineering of the BPP in *Lactuca sativa* (lettuce).

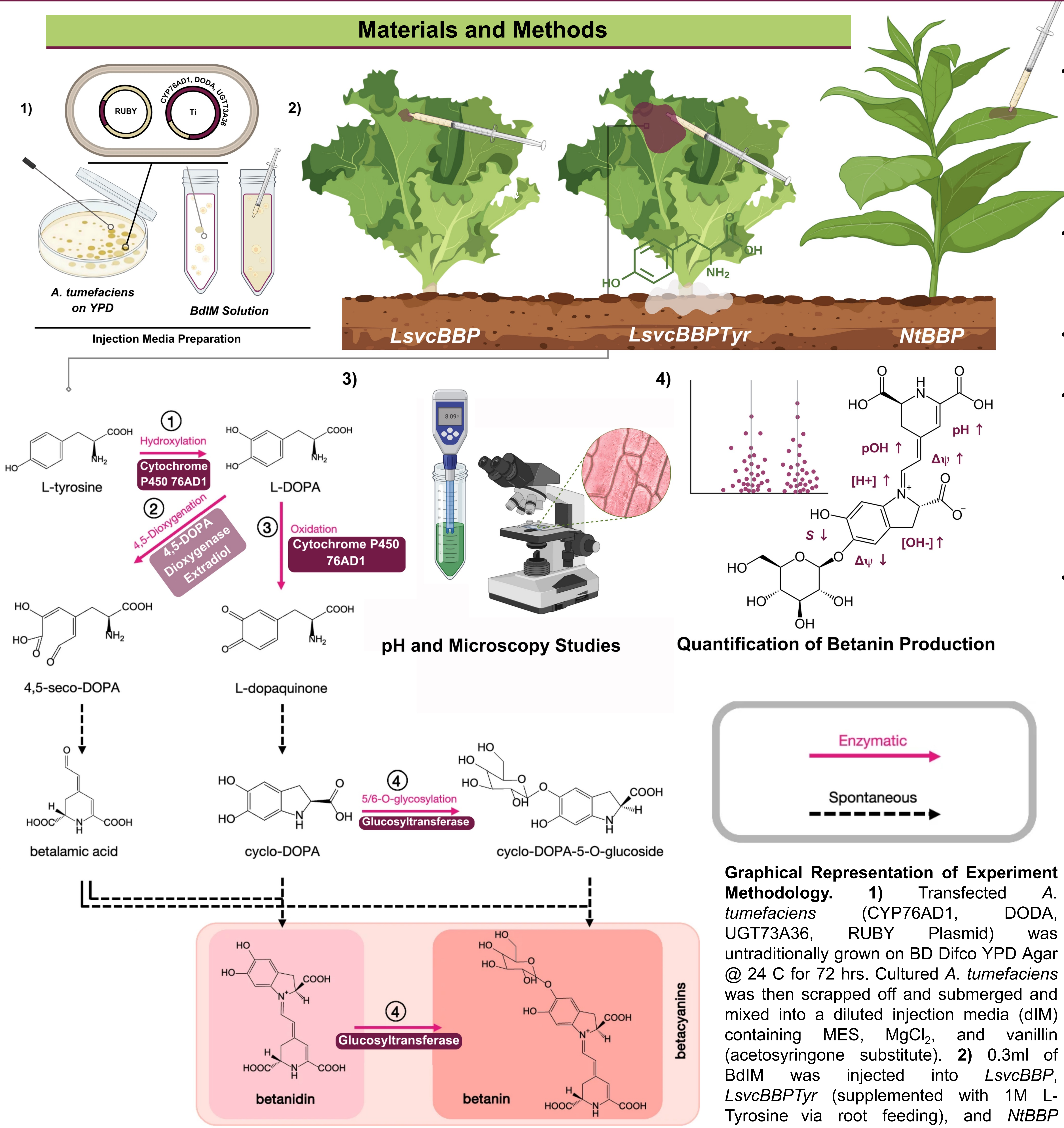
Background

While the BBP naturally occurs in species such as beets (*Beta vulgaris*), *Amaranthus caudatus*, and certain cacti, betalains are often lost during cooking due to their hydrolytic & heat sensitivity. Consequently, many individuals miss out on their potential dietary benefits. To address this, the BBP is being reconstituted by three genes encoding the enzymes necessary for BBP mediation (CYP76AD1, DODA, and UGT73A36) in lettuce to create a nutrient-dense leafy green that retains betanin health benefits when consumed raw.

Hypothesis

I hypothesize that reconstituting the BBP through the *Agrobacterium tumefaciens*-mediated infection, incorporation, and transient expression of Cytochrome P450 76AD1, 4,5-DOPA dioxygenase extradiol, and Glucosyltransferase in *Lactuca sativa* will enable the production of betanin in raw lettuce, making it a more effective dietary source of betanin compared to heat-sensitive sources like beets. Furthermore, I hypothesize that exogenous L-Tyrosine supplementation will promote betanin biosynthesis, as betanins are aromatic indole derivatives synthesized from tyrosine.

Materials and Methods



Results and Findings

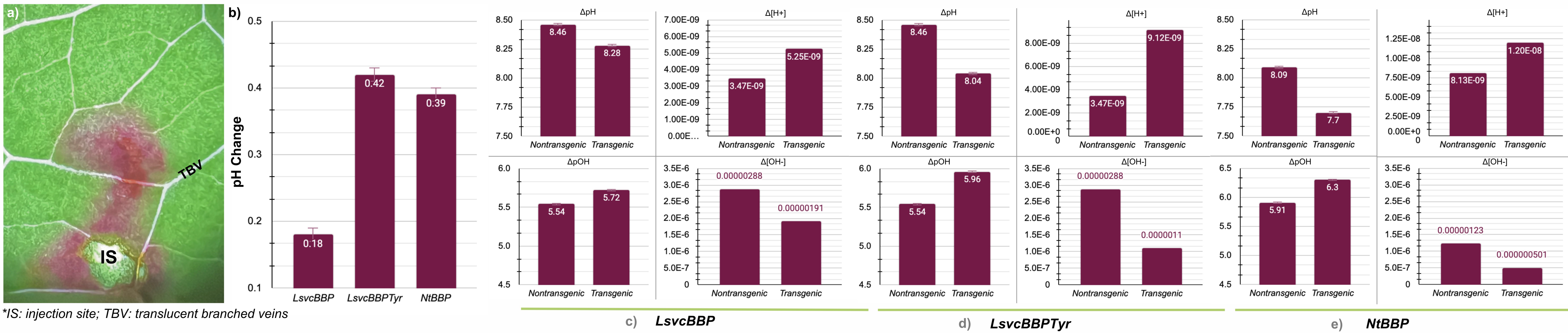
- Based on qualitative observation, *LsvcBBPTyr* produced the largest and most saturated beetroot-red-colored spots. This is consistent with the quantitative data, as the *LsvcBBPTyr* plant group experienced the greatest ΔpH .
- LsvcBBP* and *NtBBP* groups showed comparatively minimal signs of transformation, yet pH changes validated successful reconstitution of the BBP.
- Betanin production experienced a 133% increase with tyrosine supplementation.
- LsvcBBPTyr* had the greatest plant infection efficiency (relating injected bacterial volume to betanin production in terms of pH per μL of agrobacteria):
 $B2B_{LsvcBBP} = 1.32 \pm 1.08 \text{ pH}_{unit} * \mu L_{BV}^{-1}$, $B2B_{LsvcBBPTyr} = 3.08 \pm 2.52 \text{ pH}_{unit} * \mu L_{BV}^{-1}$, $B2B_{NtBBP} = 2.86 \pm 2.34 \text{ pH}_{unit} * \mu L_{BV}^{-1}$.
- The hydrolytic sensitivity of betanin is 0.016 pH units decrease for every 1% of water (by mass) increase.

Conclusions & Future Directions

This study successfully reports the **first ever betalain-biofortified lettuce GMO**, supercharging lettuce with betanin's antioxidant, anti-inflammatory, antilipidemic, antimicrobial, & anticancer properties and providing a favorable bioavailable source of high-dose betanin. This study provides the first quantification and characterization of betanin using non spectrophotometric or HPLC techniques and suggests that the enhancement of betanin biosynthesis can be achieved through ITyrosine supplementation, and in the future, the coupled supplementation of 4-Hexylresorcinol for Polyphenol Oxidase (PPO) inhibition and L-DOPA optimization.

Graphical Representation of Experiment Methodology. 1) Transfected *A. tumefaciens* (CYP76AD1, DODA, UGT73A36, RUBY Plasmid) was untraditionally grown on BD Difco YPD Agar @ 24 C for 72 hrs. Cultured *A. tumefaciens* was then scrapped off and submerged and mixed into a diluted injection media (dIM) containing MES, $MgCl_2$, and vanillin (acetosyringone substitute). 2) 0.3ml of *BdIM* was injected into *LsvcBBP*, *LsvcBBPTyr* (supplemented with 1M L-Tyrosine via root feeding), and *NtBBP* transgenic groups; 3) nontransgenic and

Data



References

- Babaei, M., Thomsen, P. T., Dyekjær, J. D., Glitz, C. U., Pastor, M. C., Gockel, P., Körner, J. D., Rago, D., & Borodina, I. (2023). Combinatorial engineering of betalain biosynthesis pathway in yeast *Saccharomyces cerevisiae*. *Biotechnology for Biofuels and Bioproducts*, 16(1). <https://doi.org/10.1186/s13068-023-02374-4>
- Selected References